Homework 5 — changed numbers (extra practice)

Module 5 - Single Oral Administration · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A single 600mg dose of medication was administered orally. The medication is about 90% orally bioavailable and is characterized as having an absorption half-life of approximately 60 minutes, elimination half-life of approximately 5 hours, and apparent volume of distribution of 30 L.

1a When does the maximum concentration of drug in the plasma occur?

1b What is the expected maximum concentration following this single oral dose?

1c What is the expected AUC following this single oral dose?

1d If the dose were increased to 1200 mg, when does the maximum concentration of drug in the plasma occur?

1e If the dose were increased to 1200 mg, what is the maximum concentration of drug in the plasma that you would expect?

1f If the dose were increased to 1200 mg, what is the AUC that you would expect?

2. A 72 kg male patient received multiple IV bolus injections of 12 mg/kg every 6 hours for 36 hours. The medication has an elimination half-life of 3 hours and an apparent

volume of distribution that is 25% of body weight.

2a What is the average concentration of drug in the plasma at steady state?

2b What is the maximum concentration of drug in the plasma at steady state?

2c What is the minimum concentration of drug in the plasma at steady state?

2d What is the expected concentration of drug in the plasma 8 hours after administration of the last dose?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: 2.9 hr

WHAT THE STEM GIVES

D_0	600mg	*not needed: t_{max} does not depend on the dose*
F	90%	*not needed: F scales height, not timing*
absorption t½	approximately 60 minutes	1 hour, so $k_a = \frac{0.693}{1} = 0.6930\ hr^{-1}$
t½	approximately 5 hours	gives $k = \frac{0.693}{5} = 0.1386\ hr^{-1}$
V_D	30 L	*not needed: t_{max} has no volume in it*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_a = \frac{0.693}{1\ hr} = 0.6930\ hr^{-1};\ k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$$

The absorption half-life of 60 minutes is 1 hour, so k_a, the absorption rate constant, is $\frac{0.693}{1\ hr} = 0.6930\ hr^{-1}$, and k, the elimination rate constant, is $\frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$. Each is 0.693 over its own half-life because $k\ t_{½} = \ln 2$; both are in reciprocal hours.

ALGEBRA

$$t_{max} = \frac{\ln(0.6930/0.1386)}{(0.6930 - 0.1386)\ hr^{-1}} = \frac{1.6094}{0.5544\ hr^{-1}} = 2.90\ hr$$

t_{max}, the time of the peak, is when the absorption rate equals the elimination rate: $t_{max} = \frac{\ln(k_a/k)}{k_a - k} \cdot \frac{0.6930}{0.1386} = 5.0$, a pure ratio of rate constants, and $\ln 5.0 = 1.6094$; dividing by $(k_a - k) = 0.5544\ hr^{-1}$ inverts the reciprocal hours to hours, 2.90 hr.

> **DOES THE NUMBER MAKE SENSE?**
> Absorption, half-life 1 hour, is five times faster than elimination, half-life 5 hours, so the peak comes a few absorption half-lives in and before one elimination half-life: 2.90 hr, between 1 and 5.

Her homework wording with different numbers, so this is not the graded answer.

1b

Worked by the drill

Answer: 12.04 mg/L

WHAT THE STEM GIVES

D_0	600mg	in the numerator $F\ k_a\ D_0$

F	90%	written 0.9, in the numerator
absorption t½	approximately 60 minutes	gives $k_a = 0.6930\ hr^{-1}$, in the numerator and in $k_a - k$
t½	approximately 5 hours	gives $k = 0.1386\ hr^{-1}$, in $k_a - k$ and $e^{-k\,t_{max}}$
V_D	30 L	in the denominator $V_D(k_a - k)$

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_a = \frac{0.693}{1\ hr} = 0.6930\ hr^{-1};\ k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$$

The absorption half-life of 60 minutes is 1 hour, so k_a, the absorption rate constant, is $\frac{0.693}{1\ hr} = 0.6930\ hr^{-1}$, and k, the elimination rate constant, is $\frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$. Each is 0.693 over its own half-life because $k\ t_{½} = \ln 2$; both are in reciprocal hours.

ALGEBRA

$$C_{max} = \frac{(0.9)(600\ mg)(0.6930\ hr^{-1})}{(30\ L)(0.5544\ hr^{-1})}\left(e^{-(0.1386)(2.90)} - e^{-(0.6930)(2.90)}\right)$$

The single oral dose equation $C_p = \frac{F\,k_a\,D_0}{V_D(k_a - k)}\left(e^{-kt} - e^{-k_a\,t}\right)$ at $t = t_{max} = 2.90$ hr; F = 0.9 is the fraction absorbed, $D_0 = 600$ mg, $V_D = 30$ L. On top, $(0.9)(600)(0.6930) = 374.22\ mg \cdot hr^{-1}$; below, $(30)(0.5544) = 16.632\ L \cdot hr^{-1}$; the reciprocal hours cancel.

ROUND

$$C_{max} = (22.500)(0.5350) = 12.04\ mg/L$$

$\frac{374.22}{16.632} = 22.500$ mg/L is the coefficient. At t_{max} the elimination term $e^{-(0.1386)(2.90)} = 0.6690$ and the absorption term $e^{-(0.6930)(2.90)} = 0.1340$; their difference, 0.5350, times the coefficient gives $C_{max} = 12.04$ mg/L.

> **DOES THE NUMBER MAKE SENSE?**
>
> If the whole absorbed dose were in the body at once the level would be $\frac{0.9 \times 600}{30} = 18$ mg/L; some has been eliminated by 2.90 hr and a little is unabsorbed, so C_{max} is below 18: 12.04 mg/L.

Her homework wording with different numbers, so this is not the graded answer.

1c

Worked by the drill

Answer: 129.9 (mg/L)·hr

WHAT THE STEM GIVES

D_0	600mg	in the numerator F D_0
F	90%	written 0.9, in the numerator
absorption t½	approximately 60 minutes	*not needed: AUC does not depend on k_a*

t½	approximately 5 hours	gives k = 0.1386 hr^{-1}, in the denominator k V_D
V_D	30 L	in the denominator k V_D

THE WORKING, ONE LINE AT A TIME

SETUP

$$AUC = \frac{F\ D_0}{k\ V_D} = \frac{(0.9)(600\ mg)}{(0.1386\ hr^{-1})(30\ L)} = 129.9\ (mg/L)\cdot hr$$

AUC, the area under the concentration–time curve, is the amount reaching the circulation over the clearance, because Cl × AUC = F D_0. Only the absorbed F = 0.9 of the 600 mg counts; Cl = k × V_D = (0.1386 hr^{-1})(30 L) = 4.1580 L/hr, and mg over L/hr is (mg/L)·hr.

> **DOES THE NUMBER MAKE SENSE?**
> AUC is the absorbed dose over clearance; 0.9 × 600 mg over (0.1386 hr^{-1})(30 L) comes to about 130 (mg/L)·hr: 129.9. Doubling the dose would double it.

Her homework wording with different numbers, so this is not the graded answer.

1d Worked by the drill

Answer: 2.9 hr

WHAT THE STEM GIVES

D_0	1200 mg	*not needed: t_{max} has no dose in it*
F	90%	*not needed*
absorption t½	approximately 60 minutes	gives k_a = 0.6930 hr^{-1}
t½	approximately 5 hours	gives k = 0.1386 hr^{-1}
V_D	30 L	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max} = \frac{\ln(k_a/k)}{k_a - k} = 2.90\ hr,\ unchanged$$

$t_{max} = \frac{\ln(k_a/k)}{k_a - k}$ contains only k_a = 0.6930 hr^{-1}, the absorption rate constant, and k = 0.1386 hr^{-1}, the elimination rate constant; the dose D_0 does not appear. Doubling the dose to 1200 mg changes neither constant, because both processes are first order, so the peak still comes at 2.90 hr.

> **DOES THE NUMBER MAKE SENSE?**
> Both absorption and elimination are first order, so doubling the dose doubles every concentration but moves no timing: the peak is still at 2.90 hr, the part a value.

Her homework wording with different numbers, so this is not the graded answer.

1e Worked by the drill

Answer: 24.07 mg/L

WHAT THE STEM GIVES

D_0	1200 mg	twice 600 mg, so twice the part b peak
F	90%	unchanged
absorption t½	approximately 60 minutes	unchanged, so t_{max} is unchanged
t½	approximately 5 hours	unchanged
V_D	30 L	unchanged

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{max} = 2 \times 12.04 \text{ mg/L} = 24.07 \text{ mg/L}$

With first-order kinetics the whole curve scales with the dose, so twice the dose gives twice the peak at the same time.

> **DOES THE NUMBER MAKE SENSE?**
> C_{max} is proportional to the dose, so twice 600 mg gives twice 12.04 mg/L: 24.07 mg/L, above the part b peak.

Her homework wording with different numbers, so this is not the graded answer.

1f Worked by the drill

Answer: 259.7 (mg/L)·hr

WHAT THE STEM GIVES

D_0	1200 mg	twice 600 mg, so twice the part c area
F	90%	unchanged
absorption t½	approximately 60 minutes	*not needed: AUC does not depend on k_a*
t½	approximately 5 hours	unchanged
V_D	30 L	unchanged

THE WORKING, ONE LINE AT A TIME

SETUP

$\text{AUC} = 2 \times 129.9 = 259.7 \text{ (mg/L)·hr}$

$\text{AUC} = \frac{F\,D_0}{Cl}$: the amount absorbed over the clearance. F = 0.9 and $Cl = k \times V_D = 4.1580$ L/hr do not change with the dose under first-order kinetics, so AUC is proportional to D_0; 1200 mg is twice 600 mg, and the area doubles from 129.9 to 259.7 (mg/L)·hr.

> **DOES THE NUMBER MAKE SENSE?**

AUC is proportional to the dose at a fixed clearance, so twice 600 mg gives twice 129.9: 259.7 (mg/L)·hr, above the part c area.

Her homework wording with different numbers, so this is not the graded answer.

2a

Worked by the drill

Answer: 34.63 mg/L

WHAT THE STEM GIVES

weight	72 kg	scales both the dose and the volume
D_0	12 mg/kg	scaled to 72 kg: 864 mg, the numerator
τ	every 6 hours	in the denominator V_D k τ
duration	36 hours	*not needed for the number: 12 half-lives, so steady state is reached*
t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	25% of body weight	scaled to 72 kg: 18 L, in the denominator

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (12\ mg/kg)(72\ kg) = 864\ mg;\ V_D = (0.25)(72\ kg) = 18\ L;\ k = \frac{0.693}{3} = 0.231\ hr^{-1}$$

The dose is 12 mg per kilogram and the volume of distribution V_D is 25% of body weight, so both are multiplied by the 72 kg weight: D_0 = 864 mg and V_D = 18 L, the kilograms cancelling. k, the elimination rate constant, is $\frac{0.693}{t½} = \frac{0.693}{3\ hr} = 0.231\ hr^{-1}$.

ALGEBRA

$$C_{avg}^{\infty} = \frac{D_0}{V_D\ k\ \tau} = \frac{864\ mg}{(18\ L)(0.231\ hr^{-1})(6\ hr)} = 34.63\ mg/L$$

C_{avg}^{∞}, the average steady-state concentration, is the dose absorbed per interval over the plasma volume cleared per interval, Cl × τ with Cl = k × V_D: (18 L)(0.231 hr^{-1})(6 hr) = 24.95 L, the hours cancelling. $\frac{864\ mg}{24.95\ L} = 34.63$ mg/L; F, the fraction absorbed, is 1 for an IV dose.

DOES THE NUMBER MAKE SENSE?

$C_0 = \frac{864}{18}$ mg/L per dose and kτ = 0.231 × 6, so the average is C_0 over kτ, a little under three-quarters of C_0: 34.63 mg/L, between the steady-state trough and peak.

Her homework wording with different numbers, so this is not the graded answer.

2b

Worked by the drill

Answer: 64.01 mg/L

WHAT THE STEM GIVES

weight	72 kg	scales both the dose and the volume
D_0	12 mg/kg	scaled to 72 kg: 864 mg, the numerator of C_0
τ	every 6 hours	the exponent $k\tau = 0.231 \times 6 = 1.386$
duration	36 hours	*not needed for the number: 12 half-lives, so steady state is reached*
t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	25% of body weight	scaled to 72 kg: 18 L, the denominator of C_0

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (12\ mg/kg)(72\ kg) = 864\ mg;\ V_D = (0.25)(72\ kg) = 18\ L;\ k = \frac{0.693}{3} = 0.231\ hr^{-1}$$

The dose is 12 mg per kilogram and the volume of distribution V_D is 25% of body weight, so both are multiplied by the 72 kg weight: $D_0 = 864$ mg and $V_D = 18$ L, the kilograms cancelling. k, the elimination rate constant, is $\frac{0.693}{t½} = \frac{0.693}{3\ hr} = 0.231\ hr^{-1}$.

ALGEBRA

$$C_0 = \frac{864\ mg}{18\ L} = 48\ mg/L;\ k\tau = (0.231)(6) = 1.386;\ e^{-k\tau} = 0.2501$$

$C_0 = \frac{D_0}{V_D} = 48$ mg/L is the rise in concentration each dose produces. $k\tau = (0.231\ hr^{-1})(6\ hr) = 1.386$, a pure number, and $e^{-k\tau} = 0.2501$ is the fraction of each dose left when the next is given; the interval $\tau = 6$ hr is 2 half-lives, so a quarter remains.

ALGEBRA

$$C_{max}^{\infty} = \frac{48\ mg/L}{1 - 0.2501} = 64.01\ mg/L$$

Each dose adds $C_0 = 48$ mg/L to what remains of the earlier doses; at steady state the peak is the sum $C_0(1 + e^{-k\tau} + e^{-2k\tau} + \ldots) = \frac{C_0}{1 - e^{-k\tau}}$. $1 - 0.2501 = 0.7499$, so $C_{max}^{\infty} = \frac{48\ mg/L}{0.7499} = 64.01$ mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is two half-lives of 3 hours, so a quarter of each dose remains at the next; the peak is $C_0 = 48$ mg/L over $1 - 0.2501$, a third more than 48: 64.01 mg/L, between 48 and 2×48.

Her homework wording with different numbers, so this is not the graded answer.

2c

Worked by the drill

Answer: 16.01 mg/L

WHAT THE STEM GIVES

weight	72 kg	scales the dose and the volume

D_0	12 mg/kg	scaled to 864 mg; C_0 = 48 mg/L, the numerator of the trough line
τ	every 6 hours	the exponent kτ, with $e^{-k\tau} = 0.2501$
duration	36 hours	*not needed for the number: steady state is reached*
t½	3 hours	gives k = 0.231 hr^{-1}; 6 hours is two half-lives
V_D	25% of body weight	scaled to 18 L, the denominator of C_0

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (12\ \text{mg/kg})(72\ \text{kg}) = 864\ \text{mg};\ V_D = 0.25 \times 72\ \text{kg} = 18\ \text{L};\ C_0 = \frac{864\ \text{mg}}{18\ \text{L}} = 48\ \text{mg/L}$$

The dose and the volume both scale with the 72 kg weight, so C_0, the rise in concentration each dose produces, is the dose over the volume.

SETUP

$$k = \frac{0.693}{3\ \text{hr}} = 0.231\ \text{hr}^{-1};\ k\tau = 0.231 \times 6 = 1.386;\ e^{-k\tau} = 0.2501$$

The printed trough line wants k and τ: k from the 3-hour half-life, and the 6-hour interval gives kτ, a pure number.

ALGEBRA

$$C_{min}^{\infty} = \frac{C_0\, e^{-k\tau}}{1 - e^{-k\tau}} = \frac{(48\ \text{mg/L})(0.2501)}{1 - 0.2501} = 16.01\ \text{mg/L}$$

This is the printed trough line, read directly with C_0 from the stem: no peak is needed. Part b reaches the same number as the 64.01 mg/L peak times $e^{-k\tau}$.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is two half-lives of 3 hours, so a quarter of each dose is left when the next goes in; the trough is C_0 times 0.2501 over 1 − 0.2501, a third of the 48 mg/L C_0: near 16.01 mg/L, and below C_0.

Her homework wording with different numbers, so this is not the graded answer.

2d

Worked by the drill

Answer: 10.08 mg/L

WHAT THE STEM GIVES

weight	72 kg	scales dose and volume inside the part b peak
D_0	12 mg/kg	inside C_{max}^{∞} = 64.01 mg/L from part b, the level the decay starts from

τ	every 6 hours	*not needed: the time asked is counted from the last dose*
duration	36 hours	*not needed for the number: steady state is reached*
t½	3 hours	gives $k = 0.231\ hr^{-1}$ for the exponent $kt = 0.231 \times 8$
V_D	25% of body weight	inside the part b peak
t	8 hours after administration of the last dose	t in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$$C = C_{max}^{\infty}\, e^{-kt} = (64.01\ mg/L)\, e^{-(0.231)(8)} = (64.01)(0.1576) = 10.08\ mg/L$$

After the last dose nothing more is given, so the level follows $C = C_0\, e^{-kt}$ from the steady-state peak $C_{max}^{\infty} = 64.01$ mg/L, the concentration the moment the last dose is in. $kt = (0.231\ hr^{-1})(8\ hr) = 1.848$, a pure number, and $e^{-1.848} = 0.1576$ of the peak remains after 8 hours.

> **DOES THE NUMBER MAKE SENSE?**
> Eight hours is between two and three half-lives of 3 hours, so between an eighth and a quarter of the 64.01 mg/L peak remains: 10.08 mg/L, between 0.125×64.01 and 0.25×64.01.

Her homework wording with different numbers, so this is not the graded answer.